

Bacterial Surface Markers in Blood: Evidence Dossier

This dossier summarizes the top bacterial-specific surface markers that are most reliable, clinically validated, and detectable in blood samples for diagnostic purposes, particularly in sepsis research.

Lipopolysaccharide (LPS) – Endotoxin (Gram-negative bacteria)

- Gold-standard marker of Gram-negative bacteremia and sepsis.
- Clinically measurable using Limulus Amebocyte Lysate (LAL) assay and ELISA.
- Elevated plasma LPS levels correlate with severity and mortality in septic patients.

Reference: Opal SM et al., Clin Microbiol Rev. 2014;27(2):246-276.

Lipoteichoic Acid (LTA) – Gram-positive bacteria

- Surface glycolipid of Gram-positive bacteria such as Staphylococcus and Streptococcus.
- Detectable in plasma during bacteremia using ELISA and immunoassays.
- Used experimentally and clinically as a sepsis biomarker.

Reference: Wartha F, Henriques-Normark B. Microbes Infect. 2008;10(9):837-844.

Peptidoglycan (PGN) Fragments

- Conserved cell wall component shed into circulation during infection.
- Detectable by specific ELISA and mass spectrometry.
- Recognized as a pathogen-associated molecular pattern (PAMP) activating immune response.

Reference: Bone RC. Chest. 1996;109(2):319-329.

Flagellin (FliC protein)

- Conserved structural protein of bacterial flagella, released into blood during systemic infections.
- Detectable by ELISA and TLR5-based biosensors.
- Elevated levels associated with bacteremia and septic shock.

Reference: Bloes DA et al., Front Immunol. 2017;8:275.

CpG Bacterial DNA (Unmethylated CpG motifs)

- Released into blood during bacteremia and sepsis.
- Detectable using PCR-based assays.
- Acts as a broad biomarker distinguishing bacterial from viral/fungal infections.

Reference: O'Neill LA, Bowie AG. Nat Rev Immunol. 2007;7(5):353-364.

Conclusion: Among numerous bacterial surface markers, **LPS, LTA, peptidoglycan fragments, flagellin, and bacterial CpG DNA** represent the most reliable, validated, and blood-detectable candidates for diagnostic and therapeutic development in sepsis and bacteremia.